

TOPOLOGY. — *On a lemma adapted to the Tietze-Urysohn theorem.* Note^(*) by **Edmond Bonan**, presented by André Lichnerowicz.

English translation

A lemma yields a proof scarcely more elaborate than the classical proof of the separation of two closed sets in a normal space by a continuous numerical function.

Recall that a topological space E is normal if it satisfies the following axiom⁽¹⁾:

(N) *For any two disjoint closed sets A and B in E , there exist two disjoint open sets U and V such that $A \subset U$ and $B \subset V$.*

Observe at once that (N) is equivalent to the following axiom:

(N') *For any two disjoint closed sets A and B in E , there exist two open sets U and V such that $\overline{U} \cap \overline{V} = \emptyset$, $A \subset U$, $B \subset V$.*

LEMMA. — *Axiom (N) is equivalent to the following:*

(N'') *For any two countable unions of closed sets A and B , neither meeting the closure of the other, there exist two disjoint open sets U and V such that $A \subset U$ and $B \subset V$.*

Let A_i, B_i ($i = 1, 2, \dots$) be two families of closed sets, A being the union of the A_i and B that of the B_i . The conditions imposed on A and B are expressed by the relations $A_i \cap \overline{B} = \overline{A} \cap B_i = \emptyset$ for every $i = 1, 2, \dots$. By (N'), there exist two open sets U'_1 and V'_1 such that $\overline{U'_1} \cap \overline{V'_1} = \emptyset$, $A_1 \subset U'_1$, $\overline{B} \subset V'_1$. Similarly, there exist two open sets U''_1 and V''_1 such that $\overline{U''_1} \cap \overline{V''_1} = \emptyset$, $\overline{A} \subset U''_1$, $B_1 \subset V''_1$. Set $U_1 = U'_1 \cap U''_1$, $V_1 = V'_1 \cap V''_1$. These two open sets satisfy

$$\overline{U_1} \cap \overline{V_1} = \overline{U_1} \cap \overline{B} = \overline{V_1} \cap \overline{A} = \emptyset, \quad A_1 \subset U_1, \quad B_1 \subset V_1.$$

The closed sets $A_2 \cup \overline{U_1}$ and $B_2 \cup \overline{V_1}$ are contained respectively in $A \cup \overline{U_1}$ and $B \cup \overline{V_1}$, neither of these last two sets meeting the closure of the other. There exist two open sets U_2 and V_2 such that

$$\overline{U_2} \cap \overline{V_2} = \overline{U_2} \cap \overline{B} = \overline{V_2} \cap \overline{A} = \emptyset, \quad A_2 \cup \overline{U_1} \subset U_2, \quad B_2 \cup \overline{V_1} \subset V_2.$$

Continuing in this way, we define two increasing sequences of open sets U_i, V_i for $i = 1, 2, \dots$ such that

$$\overline{U_i} \cap \overline{V_i} = \overline{U_i} \cap \overline{B} = \overline{V_i} \cap \overline{A} = \emptyset, \quad A_{i+1} \cup \overline{U_i} \subset U_{i+1}, \quad B_{i+1} \cup \overline{V_i} \subset V_{i+1}.$$

The open sets U , the union of the U_i , and V , that of the V_i , are disjoint and contain A and B , respectively.

Q. E. D.

Note that, since $\overline{U} \cap V = U \cap \overline{V} = \emptyset$, $\overline{U}$ does not meet B and U does not meet $\overline{B}$.

We deduce:

THEOREM⁽³⁾. — *Axiom (N) is equivalent to the following property:*

(N''') *For any set F closed in E and any continuous mapping f defined on F with values in $[0, 1]$, there exists an extension g of f to the whole space E which is a continuous mapping of E into $[0, 1]$.*

Let P be a finite subset of the interval $[0, 1]$ with P containing 0 and 1. Suppose that for each element p of P an open set U_p has been defined, these sets satisfying: 1° $\overline{f}^{-1}([0, p]) \subset U_p$; 2° $\overline{f}^{-1}(]p, 1]) \cap \overline{U}_p = \emptyset$; 3° $\overline{U}_p \subset U_q$ for every pair p, q in P such that $0 \leq p < q \leq 1$.

For $r \in [0, 1]$ not belonging to P , let p be the supremum of the elements of P smaller than r , let q be the infimum of the elements of P greater than r , and consider the two sets

$$A = \overline{U}_p \cup \overline{f}^{-1}([0, r]) \quad \text{and} \quad B = (\mathbb{C}U_q) \cup \overline{f}^{-1}(]r, 1]).$$

A is the union of the closed sets $\overline{U}_p$ and $\overline{f}^{-1}([0, r(n-1)n^{-1}])$; B is the union of the closed sets $\mathbb{C}U_q$ and $\overline{f}^{-1}(]r(n+1)n^{-1}, 1])$ for integers n greater than $r(1-r)^{-1}$.

Since $p < r < q$, we may write successively

$$\begin{aligned} \overline{f}^{-1}([0, r]) &\subset \overline{\overline{f}^{-1}([0, r])} \subset \overline{f}^{-1}([0, r]) \subset \overline{f}^{-1}([0, q]) \subset U_q, \\ \overline{f}^{-1}(]r, 1]) &\subset \overline{\overline{f}^{-1}(]r, 1])} \subset \overline{f}^{-1}(]r, 1]) \subset \overline{f}^{-1}(]p, 1]) \subset \mathbb{C}\overline{U}_p, \\ \overline{U}_p &\subset U_q. \end{aligned}$$

These relations show that $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Apply the lemma: there exists an open set U_r such that $A \subset U_r$, $\overline{U}_r \cap B = \emptyset$, which we may express as

$$1^\circ \overline{f}^{-1}([0, r]) \subset U_r; \quad 2^\circ \overline{f}^{-1}(]r, 1]) \cap \overline{U}_r = \emptyset; \quad 3^\circ \overline{U}_p \subset U_r \subset \overline{U}_r \subset U_q.$$

Start with $U_0 = \emptyset$ and $U_1 = E$: proceeding iteratively, we construct for each rational number in the interval $[0, 1]$ an open set U_r , these sets satisfying: 1° $\overline{f}^{-1}([0, r]) \subset U_r$; 2° $\overline{f}^{-1}(]r, 1]) \cap \overline{U}_r = \emptyset$; 3° $\overline{U}_r \subset U_{r'}$ for every pair r, r' such that $0 \leq r < r' \leq 1$.

For every x in E , let $g(x)$ be the infimum of the rational numbers r such that x belongs to U_r .

For any y in F , $y \in U_r$ if $r > f(y)$ since $\overline{f}^{-1}([0, r]) \subset U_r$, and $y \notin \overline{U}_r$ if $r < f(y)$ since $\overline{f}^{-1}(]r, 1]) \cap \overline{U}_r = \emptyset$. Thus $g(y) = f(y)$.

Finally, let us show that g is continuous:

For any rational number q such that $0 < q \leq 1$, $g(x) < q$ if and only if $x \in U_r$ for some $r < q$: it follows that $\overline{g}^{-1}([0, q])$ is the union of all the open sets U_r for $r < q$, and is therefore open.

For any rational number p such that $0 \leq p < 1$, $g(x) > p$ if and only if $x \notin \overline{U}_r$ for some $r > p$: it follows that $g^{-1}(]p, 1])$ is the union of the open sets $\mathbb{C}\overline{U}_r$ for $r > p$, and is therefore open.

This completes the proof, since the sets of the form $[0, q[$ and $]p, 1]$, for rational p and q , generate the topology of $[0, 1]$.

Let us give further applications of our lemma:

PROPOSITION 1. — *In a normal space E , every subspace F which is a countable union of closed sets is normal.*

Let A and B be two disjoint closed subsets of F . In the space E , A and B satisfy the hypotheses of the lemma. There exist two open sets U and V separating them in E ; hence $U \cap F$ and $V \cap F$ are two open subsets of F separating A and B .

Recall that a subset A is a generalized $F_\sigma^{(4)}$ if, for every neighborhood U of A , there exists a countable union of closed sets A_σ such that $A \subset A_\sigma \subset U$.

PROPOSITION 2. — *Axiom (N) is equivalent to the following:*

For any two generalized F_σ subsets A and B such that $\overline{A} \cap B = A \cap \overline{B} = \emptyset$, there exist two disjoint open sets U and V such that $A \subset U$ and $B \subset V$.

The open set $\mathbb{C}\overline{B}$ (respectively $\mathbb{C}\overline{A}$) contains a countable union of closed sets A_σ (resp. B_σ) containing A (resp. B).

Set $A' = A_\sigma \cap \overline{A}$ and $B' = B_\sigma \cap \overline{B}$; A' and B' are two F_σ sets such that $\overline{A'} \cap B' = A' \cap \overline{B'} = \emptyset$. Thus there exist two open sets U and V separating them, and also A and B , since $A \subset A'$ and $B \subset B'$.

(*) Session of 4 May 1970.

(1) To which the requirement that E be Hausdorff is sometimes added, as in N. BOURBAKI, *Topologie générale*, chap. 9, p. 81.

(2) H. TIETZE, *J. de Crelle*, **145**, 1915, p. 9.

(3) P. URYSOHN, *Math. Ann.*, **94**, 1925, p. 262.

(4) Yu. SMIRNOV, *Math. Sbornik*, N.S., **29**, 1951, p. 173.

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